

314707003_3.23.R

CDH

2026-03-18

```
rm(list=ls())
library(POE5Rdata)
```

```
##
## 載入套件：'POE5Rdata'
```

```
## 下列物件被遮斷自 'package:datasets':
##
##     euro
```

```
#####
#problem3.23
alpha = 0.05
mod1 = lm(price ~ I(sqft^2), data = collegetown)
b2 = coef(mod1)[[2]]
df = df.residual(mod1)

# H0 : 40*ALPHA2 <= 13 , H1 : 40*ALPHA2 > 13
#->H0 : ALPHA2 <= 0.325 , H1 : ALPHA2 > 0.325
tc = qt(1-alpha, df)
smod1 = summary(mod1)
se_b2 = coef(smod1)[2,2]
t_star = (b2 - 0.325)/se_b2

cat(
  "3.23(a)", "\n",
  "test statistic is: t = (alpha2_hat - 0.325)/se(alpha2_hat)", "\n",
  "rejection region: tc > ", tc, "\n",
  "test p-value : ", t_star, "\n",
  "hence t* is falls in the non-rejection region", "\n",
  sep=""
)
```

```
## 3.23(a)
## test statistic is: t = (alpha2_hat - 0.325)/se(alpha2_hat)
## rejection region: tc > 1.647919
## test p-value : -26.72875
## hence t* is falls in the non-rejection region
```

```
#3.23(b)

# H0 : 80*ALPHA2 <= 13 , H1 : 80*ALPHA2 > 13
#->H0 : ALPHA2 <= 0.1625 , H1 : ALPHA2 > 0.1625
```

```

t_star_b = (b2 - 0.1625)/se_b2

cat(
  "3.23(b)", "\n",
  "test statistic is: t = (alpha2_hat - 0.1625)/se(alpha2_hat)", "\n",
  "rejection region: tc > ", tc, "\n",
  "test p-value : ", t_star_b, "\n",
  "hence t* is falls in the non-rejection region", "\n",
  sep=""
)

```

```

## 3.23(b)
## test statistic is: t = (alpha2_hat - 0.1625)/se(alpha2_hat)
## rejection region: tc > 1.647919
## test p-value : 4.189467
## hence t* is falls in the non-rejection region

```

```

#3.23(c)
# E(price|sqft=20)_hat = alpha1_hat + 400 * alpha2_hat
b1 = coef(mod1)[[1]]
e_hat = b1 + 400 * b2
se_l = sqrt(vcov(mod1)[1,1] + 160000*vcov(mod1)[2, 2] + 800*vcov(mod1)[1,2])
tc_c = qt(1-alpha/2, df)
lowb = e_hat - tc_c*se_l
upb = e_hat + tc_c*se_l

cat(
  "3.23(c)", "\n",
  "expected price is ", e_hat, "\n",
  "95% CI of the expected price: [", lowb, ", ", upb, "]", "\n",
  "In repeated sampling, 95% intervals constructed this way will contain true value", "\n",
  "of expected price", "\n",
  sep=""
)

```

```

## 3.23(c)
## expected price is 167.3735
## 95% CI of the expected price: [158.0481, 176.6988]
## In repeated sampling, 95% intervals constructed this way will contain true value
## of expected price

```

```

#3.23(d)
d <- collegetown$sqft == 20
d_sqft20 <- collegetown[d, ]
s_avg = mean(d_sqft20$price)

cat(
  "3.23(d)", "\n",
  "In the sample there are 3 houses with 2000 square feet", "\n",
  "and sold ", d_sqft20$price[1], ", ", d_sqft20$price[2], ", ", d_sqft20$price[3],
  "\n",
  "The sample mean is ", s_avg, "\n",
  "and it is inside the interval we estimate", "\n",

```

```
    sep=""  
  )
```

```
## 3.23(d)  
## In the sample there are 3 houses with 2000 square feet  
## and sold 138,169,183  
## The sample mean is 163.3333  
## and it is inside the interval we estimate
```

314707003_3.31.R

CDH

2026-03-18

```
rm(list=ls())  
# install.packages('tidyr')  
library(POE5Rdata)
```

```
##  
## 載入套件：'POE5Rdata'
```

```
## 下列物件被遮斷自 'package:datasets':  
##  
##     euro
```

```
library(dplyr)
```

```
##  
## 載入套件：'dplyr'
```

```
## 下列物件被遮斷自 'package:stats':  
##  
##     filter, lag
```

```
## 下列物件被遮斷自 'package:base':  
##  
##     intersect, setdiff, setequal, union
```

```
library(tidyr)  
library(tibble)  
#####  
#problem3.31  
#(a)  
desc_table <- tuna %>%  
  summarise(  
    across(  
      c(sal1, apr1),  
      list(  
        N = ~sum(!is.na(.)),  
        mean = ~mean(., na.rm = TRUE),  
        `Std. dev.` = ~sd(., na.rm = TRUE),  
        min = ~min(., na.rm = TRUE),  
        max = ~max(., na.rm = TRUE)  
      )  
    )  
  )
```

```

) %>%
pivot_longer(
  cols = everything(),
  names_to = c("variable", ".value"),
  names_sep = "_"
)
print(as.data.frame(desc_table))

```

```

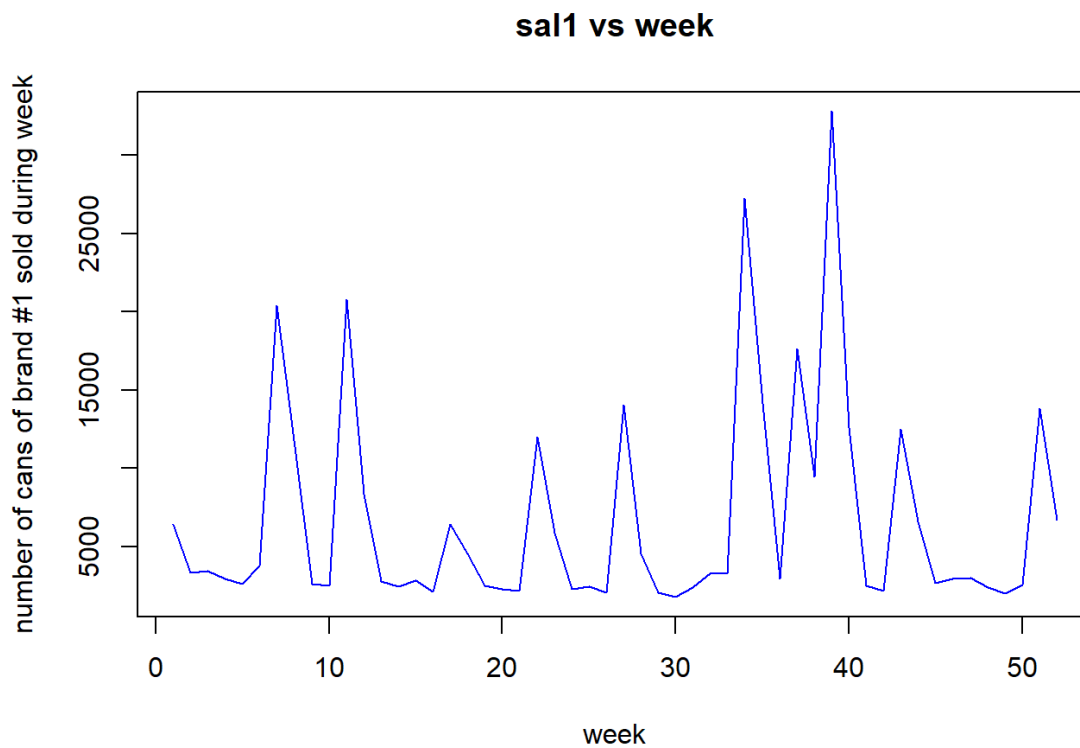
##   variable  N      mean   Std. dev.    min    max
## 1    sal1 52 6718.7115 6.915083e+03 1762.00 32820.00
## 2    apr1 52   0.7825 9.803711e-02   0.61   0.92

```

```

plot(seq(1,52), tuna$sal1,
     xlab="week",
     ylab="number of cans of brand #1 sold during week",
     main = "sal1 vs week",
     col = 'blue',
     type = "l")

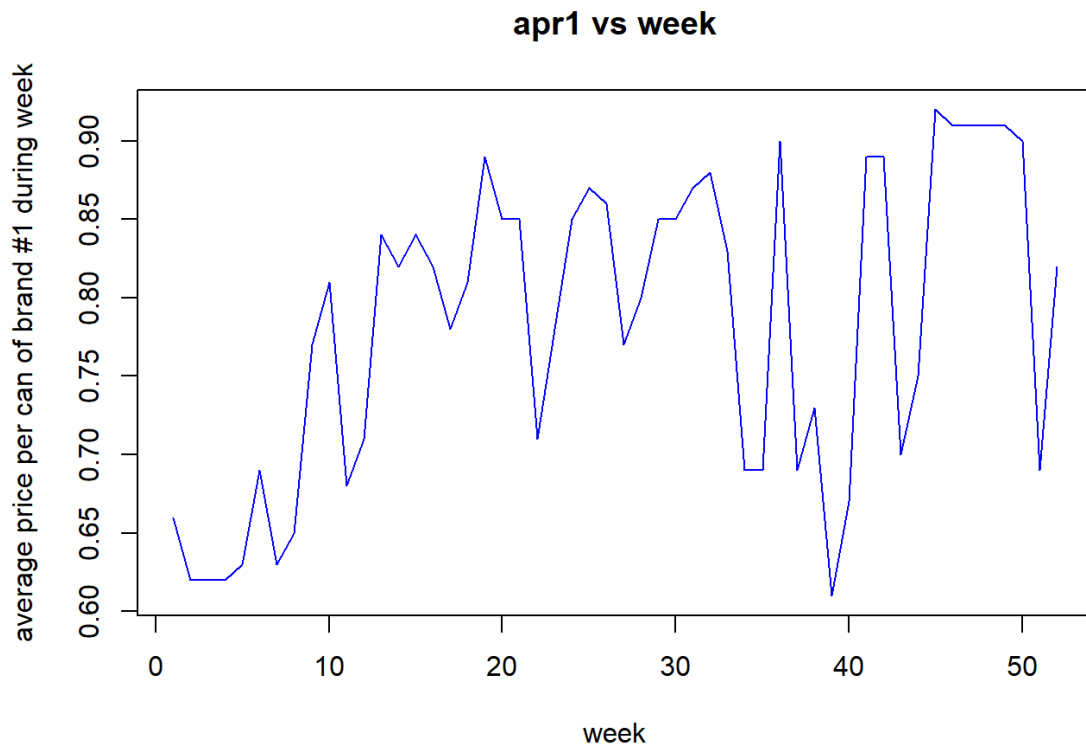
```



```

plot(seq(1,52), tuna$apr1,
     xlab="week",
     ylab="average price per can of brand #1 during week",
     main = "apr1 vs week",
     col = 'blue',
     type = "l")

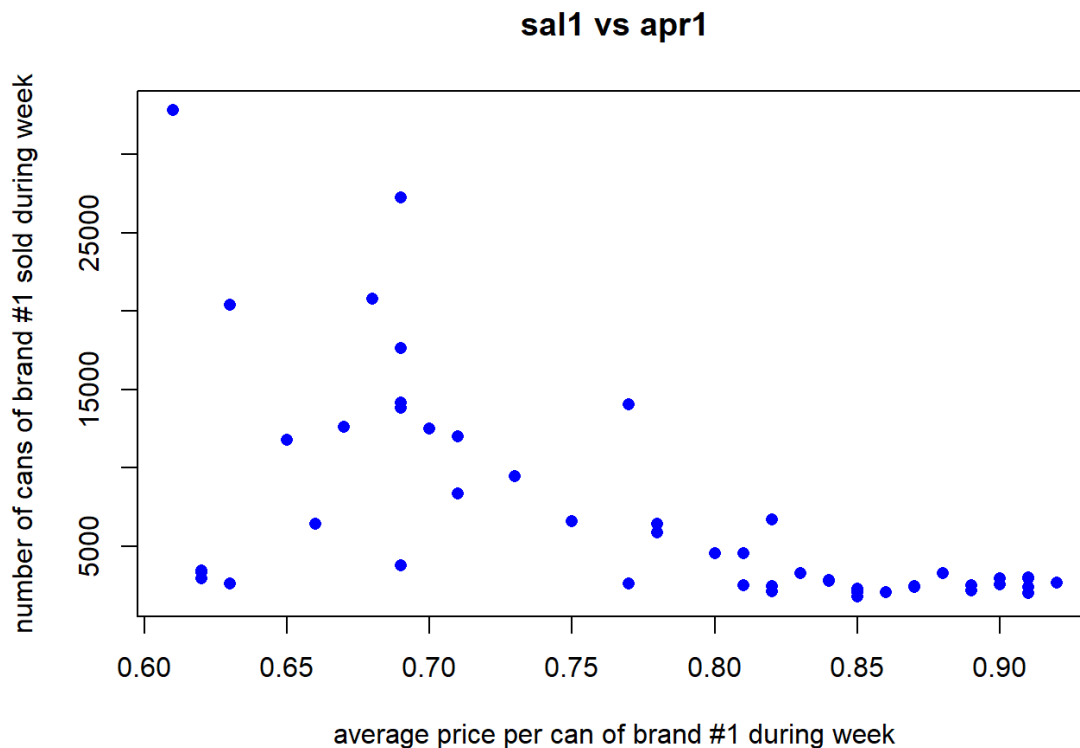
```



```
cat(
  "3.31(a)", "\n",
  "There is an inverse relationship between the weekly sales and weekly price.", "\n",
  "weekly sales exhibit much greater variation than weekly price.", "\n",
  sep = ""
)
```

```
## 3.31(a)
## There is an inverse relationship between the weekly sales and weekly price.
## weekly sales exhibit much greater variation than weekly price.
```

```
##(b)
plot(tuna$apr1, tuna$sal1,
     xlab="average price per can of brand #1 during week",
     ylab="number of cans of brand #1 sold during week",
     main = "sal1 vs apr1",
     pch = 16,col = 'blue',
     type = "p")
```



```
cat(
  "3.31(b)", "\n",
  "There is an inverse relationship between the weekly sales and weekly price.", "\n",
  "it is As expected, the higher the price, the lower the sales volume.",
  "\n",
  sep = ""
)
```

```
## 3.31(b)
## There is an inverse relationship between the weekly sales and weekly price.
## it is As expected, the higher the price, the lower the sales volume.
```

```
##(c)
price1 = 100 * tuna$apr1
mod1 = lm(tuna$sal1 ~ price1)
b1 = coef(mod1)[1]
b2 = coef(mod1)[2] #加1美分銷量減少b2罐

alpha = 0.05
df = df.residual(mod1) #degree of freedom
tc = qt(1-alpha/2, df) #critical points
smod1 = summary(mod1)
se_b2 = coef(smod1)[2,2]
lowb = b2 - tc*se_b2
upb = b2 + tc*se_b2
```

```

cat(
  "3.31(c)", "\n",
  "a one-cent increase in the price of brand one will reduce the expected",
  "\n",
  "weekly sales of brand one tuna by ", -b2, " cans", "\n",
  "and 95% CI estimate is [", lowb, ", ", upb, "]", "\n",
  sep=""
)

```

```

## 3.31(c)
## a one-cent increase in the price of brand one will reduce the expected
## weekly sales of brand one tuna by 434.4473 cans
## and 95% CI estimate is [-592.2897, -276.6049]

```

```

#(d)
# price1 = 70
e_hat = b1 + b2 * 70

alpha_d = 0.1
tc_d = qt(1-alpha_d/2, df) #critical points
se_l = sqrt(vcov(mod1)[1,1] + 4900*vcov(mod1)[2, 2] + 140*vcov(mod1)[1,2])

lowb_d = e_hat - tc_d*se_l
upb_d = e_hat + tc_d*se_l

cat(
  "3.31(d)", "\n",
  "The 90% CI estimate is [", lowb_d, ", ", upb_d, "]", "\n",
  sep=""
)

```

```

## 3.31(d)
## The 90% CI estimate is [8624.934, 11980.87]

```

```

#(e)
mod2 = lm(tuna$sal1 ~ tuna$apr1)
b2_e = coef(mod2)[2]
smod2 = summary(mod2)

alpha_e = 0.05
c = mean(tuna$apr1) / mean(tuna$sal1)
ela_hat = b2_e * c
se_ela = coef(smod2)[2,2] * abs(c)
tc_e = qt(1-alpha_e/2, df) #自由度不變所以沒改

lowb_e = ela_hat - tc_e*se_ela
upb_e = ela_hat + tc_e*se_ela

cat(
  "3.31(e)", "\n",
  "estimate of the elasticity is ", ela_hat, "\n",
  "The 95% CI estimate is [", lowb_e, ", ", upb_e, "]", "\n",
  "since absolute value of elasticity is ", abs(ela_hat), "\n",
  "this is elastic and hence the result satisfied Law of Demand and Supply",

```



```
"\n",  
  sep=""  
)
```

```
## 3.31(e)  
## estimate of the elasticity is -5.059825  
## The 95% CI estimate is [-6.898149,-3.221501]  
## since absolute value of elasticity is 5.059825  
## this is elastic and hence the result satisfied Law of Demand and Supply
```

```
##(f)  
# test statistic is (ela_hat + 3)/ (c *se(b1))  
alpha_e = 0.1  
t_test = (ela_hat + 3) / se_ela  
tc_f = qt(1-alpha_e/2, df)  
  
# state the null and alternative hypotheses in terms of the model parameters  
# 這邊會另外寫  
#  $h_0 = 3, h_1 \neq 3$   
  
cat(  
  "3.31(f)", "\n",  
  "the rejection region is  $tc > ", tc_f, "$  or  $tc < ", -tc_f, "\n",$   
  "and test p-value is ", t_test, "\n",  
  "hence We reject the null hypothesis", "\n",  
  sep=""  
)
```

```
## 3.31(f)  
## the rejection region is  $tc > 1.675905$  or  $tc < -1.675905$   
## and test p-value is -2.250572  
## hence We reject the null hypothesis
```